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Generative AI: a double-edged sword in the cyber threat landscape

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Abstract
Generative AI's swift progress advances onto profound cybersecurity dilemmas. Its usage by malevolent entities to automate intricate malware creation poses a significant threat, circumventing conventional defensive measures. This paradigm shift enables the generation of polymorphic malware, eluding signature-based detection and facilitating precision-targeted assaults. The democratization of Generative AI exacerbates these threats by extending advanced capabilities to a broader spectrum of malicious actors. A comprehensive examination of AI-generated malware's prevalence, and its repercussions is imperative to fortify cyber resilience. Such scrutiny informs proactive defense strategies vital for safeguarding digital assets within increasingly interconnected systems. Robust threat intelligence frameworks and AI-centric defensive mechanisms emerge as imperative shields against evolving cyber perils. Addressing this emergent challenge stands as an indispensable endeavor in contemporary cybersecurity discourse.

Keywords Generative AI · Cyber threats · GenAI malware · Evasion · Detection · Cyber space · ChatGPT · BARD.

1 Introduction

Generative Artificial Intelligence (GenAI) refers to advanced computational methods designed to create seemingly new and meaningful content, such as text, images, or audio, based on training data. With examples like DALL-E 2, GPT-4, and Copilot, this technology is transforming how we work and interact, ushering in a new era of innovation and communication (Martín 2024). GenAI has rapidly evolved since the 1990s, with milestones including the introduction of early generative models like Restricted Boltzmann Machines (RBMs), and Variational Autoencoders (VAEs) (Cao et al. 2023). The late 2010s saw the

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development of extensions like Conditional GANs and Progressive GANs, addressing limitations and improving performance (Showrov et al. 2024). However, the breakthrough came in 2014 with Generative Adversarial Networks (GANs), which revolutionized generative modeling by pitting a generator against a discriminator in a minimax game. In the 2020s, the rise of transformer architectures, particularly in language models such as Google's BERT and OpenAI's GPT series, marked a paradigm shift. Today, GenAI models find diverse applications in art, content creation, healthcare, and beyond, but ethical considerations regarding bias, privacy, and misuse persist. Despite challenges, the trajectory of GenAI reflects continuous innovation, driven by the advancements in deep learning and computational resources, shaping the future of artificial intelligence and human–machine interactions (Ferrara 2023).

In this dynamic landscape, the role of GenAI in cybersecurity is bi-faceted. While, it introduces new risks and challenges, it also presents opportunities for innovation and resilience. By embracing GenAI as a strategic asset and investing in research and development, organizations can leverage its potential to strengthen cybersecurity postures, mitigate risks, and adapt to the evolving threat landscape (Anandita Iyer and Umadevi 2023). Ultimately, whether GenAI is perceived as a threat or a blessing in cybersecurity depends on how effectively it is wielded to defend against emerging cyber threats while upholding privacy, security, and ethical standards (Prasad et al. 2023).

The rise of AI chatbots, as an application of GenAI, has transformed the landscape of digital communication and automation, offering numerous advantages to individuals and organizations. However, alongside these benefits emerges a complex and growing challenge. This evolving threat demands a thorough investigation to grasp the nature of malware attacks and the tactics used, determine how widespread and impactful these incidents are, and evaluate the profound consequences for people, businesses, and society at large (Barrett et al. 2023).

The lack of a deep understanding regarding the specific malware attacks planned through AI chatbots and the techniques deployed by cybercriminals creates a vast risk to digital security. The absence of comprehensive case studies and prevalence data further obstructs our ability to measure the scale of this threat and its ramifications. To confront these pressing concerns and contribute to enhancing cybersecurity practices. This study aims to investigate the cyber risks associated with AI-generated malware. By examining the various attack types, prevalence rates, and impacts of such occurrences, this study highlights the widespread impact of malware generated by GenAI. These findings not only advance the comprehension of the evolving cybersecurity landscape but also offer guidance for formulating policies and procedures aimed at safeguarding individuals, businesses, and society against the dangers posed by chatbot-generated malware.

1.1 Methodology

This study reviews the dual impact of GenAI on cybersecurity, focusing on its role in threat generation and defense. Literature from 2020–2024 was sourced from IEEE Xplore, ACM, Elsevier, SpringerLink, arXiv, and industry reports using keywords like "GenAI cybersecurity" and "AI-generated malware."

Specific inclusion and exclusion criteria were applied to refine the scope of the review. Studies that focused on GenAI applications in generating malware, automating phishing

attacks, or enhancing cybersecurity defenses were prioritized. Empirical studies and systematic reviews offering technical insights or practical implications were also included. Conversely, articles lacking a direct connection to cybersecurity or those without empirical validity were excluded to maintain the rigor and relevance of the review.

Efforts were made to minimize biases by including works from various disciplines and geographical regions. However, it is acknowledged that certain limitations persist, such as the restricted availability of non-English publications and the potential focus on specific technologies or datasets. These constraints underscore the need for further exploration to provide a more holistic perspective on GenAI's implications in cybersecurity.

The rest of the paper is organized as follow: In Sect. 2 the role of GenAI is summarized in both aspects i.e., as a blessing and as a threat. In Sect. 3 emergence of the cyber threat landscape GenAI is explained, discussing attack strategies by utilizing GenAI. An extensive literature review regarding state-of-the-art work anticipating GenAI both as a defensive and offensive tool is disscussed in 4 and 5 respectively. Discussion and critical analysis is presented in Sect. 6 whereas, Sect. 7 concludes this paper.

2 Evaluating the dual impact of GenAI

GenAI presents a dichotomy in the realm of cybersecurity, serving as both a potential threat and a boon to defense mechanisms. It offers invaluable tools and techniques for strengthening cybersecurity. Organizations can enhance their threat detection capabilities, identify anomalies, and predict potential cyber-attacks with greater accuracy. Additionally, GenAI-driven systems enable proactive threat intelligence gathering, allowing cybersecurity professionals to stay ahead of emerging threats and vulnerabilities (Zhong et al. 2023). GenAI's advantageous role in cybersecurity is multifaceted as illustrated in figure 1.

- Advanced threat detection: GenAI employs deep learning algorithms, such as GANs and VAEs, to analyze vast volumes of network traffic, system logs, and user behavior data. By learning patterns and anomalies from historical data, GenAI can effectively detect and classify suspicious activities, including malware infections, unauthorized ac-

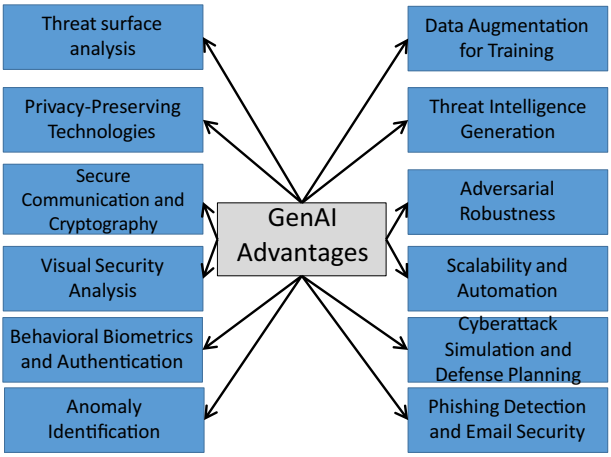

Fig. 1 GenAI Advantages

cess attempts, and anomalous behavior indicative of cyber threats (Crothers et al. 2023; Perkins et al. 2024).

- Anomaly identification: Through unsupervised learning techniques, GenAI can identify subtle deviations from normal system behavior that may signal potential security breaches. By continuously monitoring network traffic, endpoint activities, and system logs, GenAI-powered anomaly detection systems can proactively identify and respond to emerging threats, such as zero-day attacks and insider threats, before they cause significant damage (Liu et al. 2023).
- Predictive analysis: GenAI facilitates predictive analysis by leveraging machine learning models to forecast potential cyber threats and vulnerabilities based on historical data trends and emerging patterns. By analyzing contextual information, such as threat intelligence feeds, vulnerability databases, and geopolitical factors, GenAI can anticipate future attack vectors and prioritize security measures to mitigate risks effectively (Yigit et al. 2024; Gupta et al. 2023).
- Adversarial robustness: GenAI enables the development of adversarially robust defense mechanisms capable of mitigating sophisticated cyber attacks, such as adversarial examples and evasion tactics (Neupane et al. 2023). By incorporating adversarial training techniques and robust optimization algorithms, GenAI-powered systems can enhance resilience against adversarial manipulation and ensure the reliability and effectiveness of cybersecurity defenses (Qi et al. 2023).
- Scalability and automation: GenAI offers scalability and automation capabilities essential for handling the ever-increasing volume and complexity of cyber threats. By automating repetitive tasks, such as threat analysis, incident response, and vulnerability assessment, GenAI enables cybersecurity professionals to focus on strategic decision-making and proactive threat hunting, thereby improving overall operational efficiency (Huang et al. 2023).

Conversely, the sophisticated capabilities of GenAI empower malicious actors to develop increasingly complex and stealthy cyber threats. From the creation of polymorphic malware to the generation of convincing deepfake content for social engineering attacks. GenAI amplifies the scale and sophistication of cyber threats, posing significant challenges to traditional defense strategies (Gupta et al. 2023). GenAI presents a significant opportunity for malicious actors in the cyber space as well, offering advanced capabilities to conduct sophisticated and stealthy attacks (Ferrara 2023). Major threats are illustrated in figure 2.

- Automated malware generation: GenAI can be used to automate the generation of malware, including viruses, ransomware, and Trojans. By training models on large datasets of existing malware samples, malicious actors can develop algorithms capable of creating new variants that evade traditional detection mechanisms (Charfeddine et al. 2024).
- Polymorphic malware: GenAI can facilitate the creation of polymorphic malware, which dynamically changes its code structure and behavior to avoid signature-based detection. By continuously evolving in response to security measures, polymorphic malware poses a formidable challenge for antivirus and intrusion detection systems
- AI-driven phishing attacks: GenAI algorithms can be employed to generate highly convincing phishing emails, messages, or websites that mimic legitimate communications from trusted sources. By analyzing patterns in human behavior and language, malicious

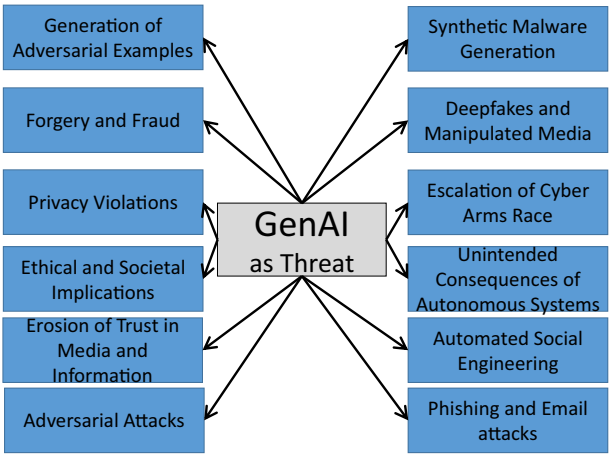

Fig. 2 GenAI as a threat

actors can craft personalized and contextually relevant phishing lures to deceive unsuspecting victims (Apruzzese et al. 2022).

- Deepfake content: GenAI enables the creation of convincing deepfake videos, audio recordings, and images that impersonate individuals or manipulate visual and auditory content. Malicious actors can use deepfake technology to spread disinformation, blackmail targets, or impersonate authority figures for social engineering purposes (Shoaib et al. 2023; Alanazi and Asif 2023).
- AI-powered social engineering: GenAI can enhance social engineering attacks by automating interactions with potential targets on social media platforms, forums, or messaging apps. By analyzing user profiles, preferences, and interactions, malicious actors can tailor persuasive messages and responses to manipulate victims into disclosing sensitive information or performing malicious actions (Schmitt and Flechais 2023).
- Adversarial attacks: GenAI can be used to generate adversarial examples, which are subtly crafted inputs designed to deceive machine learning models. By exploiting vulnerabilities in AI systems, malicious actors can bypass security controls, evade detection mechanisms, or manipulate decision-making processes to their advantage (Shashkov et al. 2023).

3 GenAI and cyber threat landscape

The threat landscape posed by GenAI technologies includes risks such as the generation of convincing deepfakes, which can lead to misinformation, identity theft, and fraud. GenAI can also be exploited to automate the creation of phishing attacks and other cybercrimes, making it easier for malicious actors to target individuals and organizations. Furthermore, GenAI can potentially generate realistic but fictitious documents, videos, or images that can be used to manipulate public opinion or sow discord. These risks necessitate robust countermeasures, such as advanced detection systems and legal frameworks, to mitigate the impact of GenAI on security and ethical concerns (Majeed and Hwang 2023).

With the use of GenAI, realistic fake content can be produced, such as deepfakes movies and artificial voices, that may be used to propagate false information or take on human iden-

tities. By propagating false information, this destroys confidence in digital content, makes fraud easier, and may have effects on politics.

Advanced GenAI technology equips adversaries with the capability to produce highly convincing fake content, encompassing images, audio, videos, and text, commonly referred to as "deepfakes". These synthetic media capabilities are made possible through the application of GANs, AutoEncoders, and other GenAI models. Threat actors exploit these techniques to create lifelike artificial faces, which can be used to populate numerous bot accounts and disseminate misinformation (Masood et al. 2023). Some popular deepfake tools are publically available in the market including Deepswap (Alanazi and Asif 2023) for face swapping, Wombo (Rancourt-Raymond and Smaili 2023) for lip-syncing, and Instagram Deep Bot for bot creation. (Xu et al. 2023).

GenAI can be used to create increasingly advance cyberattacks, such as malware that can adapt to defenses and phishing attempts. Attacks like data poisoning and model stealing can target models created by AI directly (Santoso et al. 2023).

Large datasets are frequently needed for GenAI, which raises questions concerning consent and privacy during data collection. AI able to reassemble recognizable photos or data from anonymized sources (Bendiab et al. 2023).

Concerns over authorship and originality can result from GenAI works that violate traditional intellectual property laws. The potential of AI to produce literature, music, and art raises existential concerns for those in the creative industries (Dasawat and Sharma 2023; Anandita Iyer and Umadevi 2023). In the training of data set, AI systems have the potential to reinforce and magnify societal biases. Who is accountable for the decisions and results of AI systems is a topic of continuous discussion (Taddeo et al. 2023).

AI automation also lead labor markets to shift, especially in industries like customer service and content production. The concentration of AI technology in the hands of a small number of strong companies has the potential to increase economic gaps (Sedar et al. 2023).

Current laws are not sufficient to handle particular challenges presented by GenAI. International agreements, standards and norms are necessary due to the global nature of AI technology (Saeed et al. 2023). AI-generated information is becoming more realistic, which may cause psychological consequences by obfuscating the distinction between human and machine interactions. AI overuse may have negative effects on individual's capacity to think critically and make decisions (Chamola et al. 2023). A critical debate is ongoing in scientific and creative groups regarding the use of AI for non-productive tasks and creative activities like writing and painting etc., for humans.

Emerging risks could arise from unanticipated novel uses of AI technology. The speed at which AI is developing may surpass our capacity to understand and reduce related concerns (Zambrano et al. 2023).

3.1 Major attack strategies using GenAI

Threat actors are continually enhancing their offensive capabilities through a combination of learning, collaboration, and the adoption of emerging technologies. In the past, these actors relied on manual techniques and had limited automation for launching their attacks. This involved tasks such as manually crafting phishing emails, creating malware variations by hand, conducting manual research on potential targets, and more. While these methods

had some effectiveness, they were time-consuming and demanded significant effort to scale up.

In today's landscape, threat actors have access to cutting-edge LLMs (Brown et al. 2020). These models can generate human-like language, enabling the creation of personalized and highly convincing phishing messages that closely mimic legitimate communications, which can be utilized for reconnaissance. This adaptation streamlines the labor-intensive aspects of attacks, allowing threat actors to focus on more strategic planning and enabling them to scale their operations and launch attacks more swiftly (Temara et al. 2023).

Defensive systems that are adept at recognizing and avoiding such attacks must be developed by having a thorough understanding of how they work. Without a thorough grasp of the tactics used by threat actors, creating a practical plan to reduce cyber dangers is difficult. An in-depth knowledge and awareness of these tactics enables defenders to develop strong defensive systems. An evasion approach, for example, can be used in reconnaissance to learn about a network, hide information using stenography or other techniques, or deliver a hostile payload, like malware and rootkits. Defenders also need to keep an eye out for variations of these offensive tactics. The "Living-off-the-Land" (LOTL) technique, for instance, is a type of evasion that makes use of legitimate tools and processes such as PowerShell, Windows Management Instrumentation (WMI), Mimikatz, and others to avoid detection, carry out network reconnaissance, enable lateral movement, and preserve persistence (Sharma et al. 2023). The POSHSPY assault, carried out by the Russian state-sponsored APT organization APT29, is an illustration of a LOTL-based attack (Orbinato et al. 2024). Promient attack strategies used by malicious actors focused in this work are illustrated as in figure 3. Here we identified five key attack strategies: *evasion, adaptation and automation, polymorphism, deception, and misinformation*, which are discussed in perspective of GenAI.

3.1.1 Evasion

The process of attacking while deliberately avoiding or escaping detection techniques is referred to as evasion.

Fig. 3 GenAI Attack Strategies

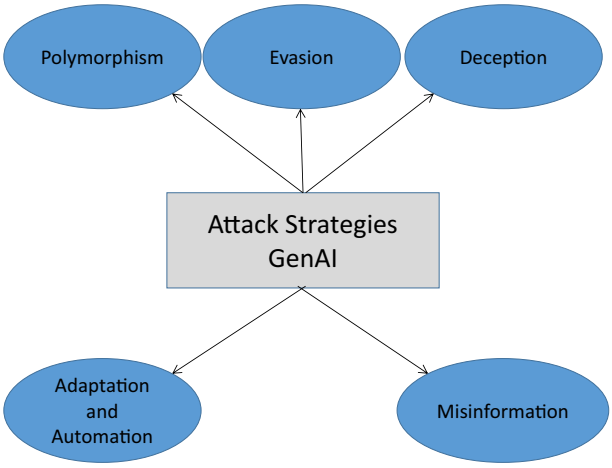

Table 1 Critical review: state of the artwork

Purpose of study	Problem addressed	Methodology	Experiments and results	Limitations/conclusion
Qammar et al. (2023) Investigates the security vulnerabilities and attacks on chatbots, particularly focusing on ChatGPT, and how malicious actors exploit these platforms.	Addresses the increasing cybersecurity challenges posed by the evolution of chatbots to sophisticated AI models like ChatGPT.	Analyzes historical chatbot developments, cybersecurity threats, and specific vulnerabilities in ChatGPT, including case studies of malicious usage.	Demonstrates the potential misuse of ChatGPT for creating malware, phishing emails, and other cyber attacks.	Recognizes the limitations in predicting and addressing all possible security threats due to the rapidly evolving nature of AI technologies.
Shoaib et al. (2023)Threats of deepfakes and misinformation in the age of advanced AI,	Use of deepfakes and misinformation poses significant challenges to global information integrity.	The analysis includes detection algorithms, AI-driven authentication methods, and collaborative strategies across platforms.	Underscores the societal impact of deepfakes and misinformation, advocating for effective detection methods,	AI advancements potentially surpassing the framework, challenges in detecting deepfakes, and global ethical cooperation difficulties.
Pa Pa et al. (2023)Auto-GPT eases the hurdle of malware generation, and existing AV and EDR detect the malware by AI.	Addresses the potential misuse of advanced AI technologies like ChatGPT in cybersecurity threats.	Creation of seven malware, (about 400 lines of code) within 90 minutes. Five AV and one EDR used to check malware in lab against virustotal, ChatGPT, text-DaVinci-003, Auto GPT used	Auto-GPT evade the safety controls detection rate of malware is below 30% detection on virustotal is 14% to16% for ransomware 9% to11% for AI-phishing mailers	Limited to the specific AI scenarios advanced attackers might innovatively utilize advanced AI.
Derner and Batistič (2023)To examine the security risks associated with ChatGPT, including its potential misuse for malicious purposes.	Addresses the challenges in ensuring the ethical and secure usage of ChatGPT in light of its capabilities for misuse.	Empirical study examining ChatGPT's content filters and potential bypass methods, along with a qualitative analysis of security implications.	Reveals that despite safeguards, ChatGPT can be manipulated to generate harmful content, posing significant security risks.	Limited by the specific version of ChatGPT tested and the evolving nature of AI and cybersecurity risks.
Schmitt and Flechais (2023)Realistic Content Creation, Advanced Targeting and Personalization, Automated Attack Infrastructure	The advancement in GenAI, presents challenges in SE and phishing	Address the integration of SE with qualitative and framework based analysis	GenAI use polymorphic attack patterns to evade detection and craft phishing emails that avoid red flags	Evolving landscape of AI
Gupta et al. (2023)Identify the vulnerabilities of ChatGPT exploited by malicious actors bypassing ethical constraints	use of GenAI tools in both the defensive and offensive side of cybersecurity	attacks like Jailbreaks, phishing attacks, SE, SQL injection Ransomware and prompt injection code generation by the ChatGPT	Successfully generation of attacks as a DAN(Do anything Now)	rapid evolution of AI technologies and the corresponding cybersecurity landscape

Table 1 (continued)

Purpose of study	Problem addressed	Methodology	Experiments and results	Limitations/conclusion
Barrett et al. (2023)Reports the findings of a workshop held at Google on the dual-use dilemma posed by GenAI	understand and mitigate the security risks associated with GenAI,	Spear Phishing Hallucinations Dissemination of Deepfakes	Significance of GenAI attacks	paper is not comprehensive and represents workshop findings, indicating ongoing evolution of GenAI risks and defenses.
Neupane et al. (2023)Examines the implications of GenAI in cybersecurity, focusing on its impact on cyber defense strategies.	Addresses the challenges of integrating GenAI in cyber defense, considering the risks and benefits.	GPT-2, GPT-3 and GPT-4 used for weaponization and exploitation	GenAI produce polymorphic, evasive, and deceptive malicious content	defensive strategies enabled by GenAI not evaluated
Esmailzadeh (2023)Examines the impact of ChatGPT in cyberspace, focusing on its role in cybersecurity and potential threats.	Addresses the dual use of ChatGPT in cybersecurity, posing both opportunities for defense and risks for exploitation.	Analyzes the functionality of ChatGPT, cybersecurity case studies, and explores potential misuse scenarios.	Highlights the versatile use of ChatGPT in creating malware, and phishing emails, and its role in cyber defense strategies.	Detection of GenAI attacks not taken into consideration.
Kaheh et al. (2023)investigate the applicability of LLMs like GPT-4 in cybersecurity, and Cyber Sentinel for streamlining security tasks.	to understand LLMs interpret cybersecurity logs and events, and take security actions based on human instructions.	creating the Cyber Sentinel system, integrating components like IoC Signature Database, a conversational agent, SIEM, and GPT-4.Explores the sequential chaining of GPT-4 models with prompt engineering across sub-tasks.	The experiments demonstrate that LLMs can be effectively used for cybersecurity tasks	the essential role of human oversight in cybersecurity tasks needs to be automated.
Alawida et al. (2023)potential cybersecurity threats posed by ChatGPT, including tactics employed by malicious actors for exploitation	The problem identified is the vulnerability of ChatGPT to malicious interactions, particularly in the context of cyber assaults.	Creation of malware for stealing data in python and java, Social Engineering and phishing, Integration of Macros and LOLBINS, survey on chatGPT	chatGPT exploitation by cyber criminals is 82%	limitations in the accuracy of responses, errors of fact detection, and algorithmic biases when ChatGPT reacts to cues
Botacin (2023) chatGPT-3 is used to generate malware, test ChatGPT to rewrite malware code in multiple manners	LLM generation capabilities might be used by attackers to assist in malware creation	C code is used as experiment language, Generate Windows APT	success rate of variants generation is below 50%, detection scores (4 to 55) by VirusTotal	results depend on the prompted queries which can further be eloborated.

Table 1 (continued)

Purpose of study	Problem addressed	Methodology	Experiments and results	Limitations/conclusion
Khoury et al. (2023)Study evaluates ChatGPT's code security, explores guided improvements, and discusses ethical implications due to its natural language-to-code translation capability.	potential insecurity of code produced by ChatGPT.	ChatGPT generates 21 programs in different languages, each showcasing vulnerabilities like SQL injection and memory corruption, without explicitly requesting secure code, simulating novice programmer behavior to evaluate its ability to create code resistant to common cyber attacks.	only 5 out of 21 use cases were secure, but ChatGPT improved security in 7 additional cases upon explicit correction requests.	specific version of ChatGPT (v. 3.5), the variability in ChatGPT's responses produce different answers to the same question, the security of the generated code often depended on the programming language used
Shashkov et al. (2023)To investigate the effectiveness of various AI and ML methods in optimizing adversarial behavior	which machine learning training methods are most effective for agent behavioral competence in cybersecurity simulations, APTs	assesses and contrasts deep reinforcement learning (DRL), ES and Monte Carlo Tree Search in three diverse simulation environments (Connect 4, SNAPT, CyberBattleSim), training agents with these methods and evaluating their performance.	in Connect 4, the (1+1)-ES algorithm performed best, while in CyberBattleSim, the A2C algorithm was effective for attackers. combination of DRL and ES algorithms	real-world scenarios are not fully established.

As an example, consider a ransomware attack such as WannaCry (Lee et al. 2023). The attackers took employ of the EternalBlue vulnerability in Microsoft Windows systems. They used evasion strategies, like obfuscation and encryption, to spread the malware quickly and evade detection. Security mechanisms like firewalls, intrusion detection and prevention systems (IPS, IDS), and malware analysis can be bypassed with the use of evasion techniques (Aslan et al. 2023).

Evasion attacks using GenAI to create forged inputs are often reported (Ferrara 2023). These inputs are intended to trick detection systems into assuming things that aren't true. Using GANs, for example, adversaries can create malicious traffic that can be used to attack intrusion detection systems, penetrate around them, and propagate throughout the network. Malicious URLs can also be created with ease to avoid being detected by URL and web detectors (AlEroud and Karabatis 2020; Apruzzese et al. 2022).

One major evasion tactic is obfuscation, which is a way for malware to hide its attack payload to avoid detection by static analysis tools and conventional anti-malware programs. Adversaries use GenAI to improve obfuscation strategies as reported in Deshpande and Gupta (2023). Sophisticated techniques such as dead code insertion, register reassignment (which involves modifying a register without affecting computer code or behavior), subroutine reordering, and more can be created.

3.1.2 Adaptation and automation

Instances of attacks employing adaptation and automation strategies include NotPetya (Neupane et al. 2023), a destructive malware that incorporated an automated propagation mechanism to spread to new systems. Another example is Crash0verride (Neupane et al. 2023), a malware framework which is designed to target Industrial Control Systems (ICS) utilized in electric power grids.

The adoption and automation of GenAI technology have led to the rise of sophisticated virtual kidnapping scams, presenting significant challenges for law enforcement and cybersecurity professionals (Habbal et al. 2024). In these schemes, attackers use AI-generated content, such as realistic audio recordings and deepfake videos, to deceive victims into believing that their loved ones are in danger (Sudhakar and Shanthi 2023). By exploiting psychological vulnerabilities and leveraging personalized social engineering tactics, perpetrators aim to extort ransom money or sensitive information from their targets (Rawat et al. 2023). Automation of these scams enables attackers to scale their operations and orchestrate complex fraud schemes with minimal manual intervention. Addressing this threat requires advanced detection algorithms, robust authentication mechanisms, increased public awareness, and international cooperation to dismantle criminal networks and prosecute perpetrators (Montasari 2024). Using technological innovations and encouraging collaboration, it is possible to mitigate the risks associated with virtual kidnapping scams enabled by GenAI and protect individuals from exploitation (Miotti and Wasil 2024; Tremont 2023).

3.1.3 Polymorphism

In general, polymorphism signifies the ability of an object or data element to assume various forms or types. In the context of network security, it denotes a strategy of self-mutation or alteration of the form of attack vectors. An illustration of the polymorphism strategy, as

reported by Darkreading, involves the SideWinder Advanced Persistent Threat (APT) group in India, which targeted Pakistani government officials and individuals. They employed polymorphism techniques to circumvent traditional signature-based antivirus detection and deliver a subsequent payload. Examples of malware utilizing this strategy include Ursnif (also known as Gozi) and AAEH (or Beebone), among others.

Polymorphism-based attacks alter malware signatures in a manner that eludes detection by anti-malware systems relying on fixed signatures, thus infiltrating the target network or systems. Presently, malware can transform into numerous polymorphic variations. Defense against polymorphic malware is approached through two methods: static analysis and dynamic analysis (Alosefer 2012). Nevertheless, static analysis can be challenging due to the dynamic nature of malware and the extensive code bases that must be scrutinized for polymorphic behavior.

Attackers can potentially exploit the capabilities of GenAI to produce synthetic data for crafting polymorphic attacks. For example, Chauhan and Heydari (2020) employed GANs to demonstrate that deep learning techniques alone were inadequate to detect new attack patterns. In their subsequent research, the same authors illustrated that attackers could launch polymorphic Distributed Denial of Service (DDoS) attacks using a generative model like WCGAN Liu (2023).

3.1.4 Deception

Deception represents an interaction between two parties—a deceiver and a target. In this interaction, the deceiver effectively convinces the target to accept a specific incorrect version of reality as true, to cause the target to act in a manner that benefits the deceiver (Rowe 2007). One instance of Trojan malware employing deceptive tactics to avoid detection is the Emotet trojan. Emotet spreads through phishing email attachments and links, which, when clicked, trigger the malware's payload. According to the MITRE ATTACK framework, Emotet employs techniques like obfuscation and software packing to elude detection.

GenAI, particularly large language models (LLMs), have formidable capabilities to generate convincing and misleading content, enabling attackers to manipulate, deceive, and exploit their targets. In the realm of offensive applications, LLMs are used for crafting deceptive content in phishing emails and social engineering messages. Attackers can create content that closely mirrors the writing style, tone, and vocabulary of specific individuals or organizations, even including pertinent details and references. This makes it challenging for recipients to discern the malicious nature of the email.

Threat actors also leverage LLMs for impersonation and identity theft. For example, they make use of readily available tools like ChatGPT (Teubner et al. 2023), Brad (Chowdhury et al. 2023), or Bing Chat (Afgiansyah 2023) to produce deceptive messages, legal documents, or official-looking notices that mimic the communication style of a trusted individual or organization. These deceptive materials are used to trick individuals into divulging sensitive information or gaining unauthorized access to network systems.

The emergence of phishing attacks via Microsoft Teams, coupled with the application of GenAI by groups like Midnight Blizzard, represents a significant escalation in cyber threats. Midnight Blizzard has leveraged Microsoft Teams as a platform to orchestrate sophisticated phishing campaigns, exploiting compromised accounts within the platform to enhance the credibility and success rate of their attacks (Bazarkina et al. 2023). By utilizing genAI, they

can craft convincing phishing messages, mimicking the writing style and communication patterns of legitimate users or organizations, further deceiving unsuspecting targets (Howard 2023). This combination of advanced technology and social engineering tactics enables Midnight Blizzard to bypass traditional security measures and manipulate users into disclosing sensitive information or performing malicious actions. Such attacks underscore the need for robust cybersecurity defenses, including user education, multi-factor authentication, and threat detection capabilities, to safeguard against evolving threats in collaborative platforms like Microsoft Teams (Ebert and Beck 2023).

3.1.5 Misinformation

Misinformation refers to false or inaccurate information that is intentionally or unintentionally created and then spread. It is essentially an attack on our cognitive understanding (Wu et al. 2019). When false information is intentionally created and propagated with the aim of misleading, it is specifically referred to as disinformation.

Threat actors, including nation-state entities driven by geopolitical interests, extremist groups, or economically motivated companies seeking to exploit information, employ the strategy of spreading misinformation. This tactic is aimed at sowing societal unrest, amplifying polarization, and in some cases, influencing election outcomes. The proliferation of social networks like Twitter (Agarwal et al. 2023), Facebook (Karlsen and Aalberg 2023), and Reddit (Arazzi et al. 2023), combined with easy internet access, has created a fertile ground for the dissemination of misinformation. These platforms have made it simple to publish content without undergoing proper editing, fact-checking, or being held accountable (Alkhair et al. 2023). While cyberattacks primarily target computer networks and end systems, deliberate misinformation and partial truths can be used to manipulate people's perspectives or divert unsuspecting users away from accessing accurate information (Hutchens 2023; Singh 2023). The convergence of pig butchering and GenAI in investment fraud schemes presents a novel and concerning trend in cybercrime. In this type of scam, perpetrators befriend victims online, often through social platforms or romance sites, establishing a sense of trust and rapport. Leveraging GenAI technology, they fabricate elaborate narratives and false evidence to lure victims into investing in fictitious cryptocurrency schemes purportedly related to pig butchering ventures (Sarkar and Shukla 2023). By exploiting the anonymity and complexity of cryptocurrency transactions, perpetrators can deceive victims into depositing their money into fraudulent schemes, promising high returns or exclusive investment opportunities. However, once the funds are transferred, the perpetrators disappear, leaving victims with significant financial losses and little recourse for recovery (Sison et al. 2023). To protect against such scams, individuals should exercise caution when engaging with strangers online, conduct thorough research before making investment decisions, and remain skeptical of unrealistic promises or high-pressure tactics.

4 State of the art: GenAI as a defensive tool

GenAI significantly enhances cybersecurity by enabling real-time threat detection, predictive analysis, and scalable automation. It improves operational efficiency and empowers security teams to focus on strategic decision-making.

GenAI revolutionizes vulnerability management by automating tasks like scanning, risk assessment, and patch prioritization, leveraging analytics to address high-risk issues effectively. Integrated with tools like SOAR platforms, GenAI automates multi-step workflows, such as detecting phishing attempts, updating firewalls, and facilitating dynamic threat hunting by aggregating and analyzing diverse data sources.These capabilities reduce manual effort, strengthen organizations' security posture, and provide a robust defense against increasingly complex cyber threats (Patil et al. 2024).

GANs detect anomalies by generating synthetic data and using a discriminator to identify deviations, such as disguised malware or irregular network activity. VAEs focus on learning and reconstructing normal system behavior, highlighting deviations that indicate potential security incidents. Together, these models enable GenAI to detect both known and novel attack vectors precisely (He et al. 2023). A key strength of GenAI is its application of unsupervised learning techniques, which allow it to identify deviations from normal system behavior without relying on predefined labels. By continuously monitoring network traffic, endpoint activities, and system logs, GenAI establishes adaptive baselines of expected behavior. These baselines serve as references to detect even minor deviations, such as unusual spikes in data transmission, connections to unfamiliar IP addresses, or atypical user activities. Such deviations are often indicative of threats like zero-day exploits, malware infections, or unauthorized access attempts (Park et al. 2022). GenAI's ability to analyze user behavior also plays a critical role in mitigating insider threats. Insiders often operate within legitimate access boundaries, making their actions difficult to detect with traditional security tools. GenAI overcomes this challenge by identifying subtle behavioral anomalies, such as accessing sensitive data at irregular hours, attempting to exfiltrate large volumes of information, or performing tasks outside their usual workflow. By flagging these activities, it empowers security teams to respond proactively, often neutralizing threats before significant damage occurs (Mavikumbure et al. 2024). Furthermore, GenAI enhances endpoint security by driving adaptive, behavior-based monitoring tools. These tools detect abnormal process executions, unauthorized file access, and deviations in application behavior. Its integration with dynamic threat intelligence platforms enables organizations to prioritize responses, visualize attack patterns, and predict potential vulnerabilities, thus strengthening their overall cybersecurity posture (Liu et al. 2023). The proactive nature of GenAI-driven detection ensures timely identification and mitigation of advanced threats, such as zero-day attacks and insider exploits. While its transformative potential is undeniable, it is imperative to address associated risks, such as the misuse of GenAI by adversaries for crafting sophisticated attacks. Balancing innovation with robust ethical and regulatory measures will be crucial in harnessing the full potential of GenAI for securing the digital ecosystem (Crothers et al. 2023; Perkins et al. 2024). By utilizing advanced machine learning models, GenAI goes beyond traditional predictive techniques to forecast emerging cyber threats based on historical data trends, real-time contextual information, and evolving patterns in the threat landscape (Duary et al. 2024). GenAI employs supervised and unsupervised learning models, along with deep neural networks, to analyze diverse datasets, including threat intelligence feeds, vulnerability databases, and system activity logs. For example, a model trained on vulnerability databases, such as the Common Vulnerabilities and Exposures (CVE) repository, can learn patterns associated with past exploits. When combined with real-time threat intelligence feeds, such as Indicators of Compromise (IOCs) and adversary tactics from frameworks like MITRE ATT&CK, GenAI identifies indicators

that may predict future attacks (Subbaratinam 2022). One of the key advantages of GenAI in predictive analysis is its ability to prioritize security measures based on risk assessment. GenAI models calculate the probability of specific attack vectors and the potential impact of successful exploits, enabling organizations to focus on mitigating the most critical risks (Khalil 2024). Integrating GenAI with predictive cybersecurity tools requires robust architectures capable of handling large-scale data processing. Distributed computing frameworks like Apache Spark or TensorFlow are commonly used to train and deploy GenAI models. To ensure real-time analysis, organizations often implement these systems in cloud-based environments equipped with GPU-accelerated infrastructure, enabling faster computation and scalability. By combining historical insights, real-time data, and context-aware predictions, GenAI empowers organizations to anticipate cyber threats with unprecedented precision. This proactive approach not only enhances threat detection and prevention but also reduces response times, minimizing the potential impact of sophisticated cyberattacks. GenAI introduces unparalleled scalability and automation capabilities to the field of cybersecurity, addressing the growing volume and sophistication of cyber threats. Its ability to process vast amounts of data and execute complex tasks autonomously is transforming how organizations detect, respond to, and mitigate cyber risks. These capabilities are especially critical in modern cybersecurity operations, where the sheer volume of potential threats often overwhelms human analysts. GenAI enables scalability by leveraging advanced computational frameworks and distributed processing systems. Using platforms like Apache Spark, Kubernetes, or cloud-based infrastructure with GPU acceleration, GenAI can analyze petabytes of data across multiple sources, including network logs, endpoint telemetry, and threat intelligence feeds, in real-time. This capability ensures that organizations can scale their security operations to match the exponential growth of cyberattack vectors without compromising performance (Paulose and Neelanath 2024). In threat analysis, GenAI models can automatically classify and prioritize alerts generated by Security Information and Event Management (SIEM) systems. By learning from historical attack data and leveraging real-time threat feeds, GenAI can distinguish between false positives and genuine threats, drastically reducing alert fatigue for security teams (Ali et al. 2024). In incident response, GenAI-powered systems automate critical processes, such as the identification of compromised assets, containment of threats, and remediation of vulnerabilities. For example, a GenAI model integrated with an Endpoint Detection and Response (EDR) platform can detect malware, isolate affected endpoints, and initiate recovery procedures without requiring manual intervention. These systems also use machine learning algorithms to learn from past incidents, continuously refining their response strategies and adapting to new attack patterns (Grigorev et al. 2024).

5 State of the art: GenAI as an offensive tool

In Shashkov et al. (2023) the focus of research was to understand which machine learning training methods are most effective for agent behavioral competence particularly in the context of Advanced Persistent Threats (APTs). The study reveals that the combination of DRL and ES algorithms can lead to effective adversarial strategies in cybersecurity simulations.

In Schmitt and Flechais (2023), authors addressed the GenAI based attacks regarding the realms of SE and phishing, necessitating an understanding of these evolving threats. The

research employed a mixed-methods approach, including a systematic literature review, AI capabilities analysis, and development of the GenAI SE Framework.

In Barrett et al. (2023), authors investigated the security risks posed by GenAI technologies, including their dual use for both beneficial and malicious purposes.

In Neupane et al. (2023), authors examined the implications of GenAI in cybersecurity, focusing on its impact on cyber defense strategies. In Pa Pa et al. (2023), authors investigated the capabilities of ChatGPT and similar AI models in generating functional malware and attack tools. This research addressed the potential misuse of advanced AI technologies like ChatGPT in cybersecurity threats. Authors have conducted experiments to develop malware using ChatGPT and Auto-GPT, testing their functionality and detection by security solutions. Results demonstrates that ChatGPT can be used to create various types of malware, which are often not detected by standard security products. The study conducted in Derner and Batistič (2023) investigated the security risks associated with ChatGPT, including its potential misuse for malicious purposes. Addressed the challenges in ensuring the ethical and secure usage of ChatGPT in light of its capabilities for misuse. Conducted an empirical study looking at ChatGPT's content filters and potential bypass methods, along with a qualitative analysis of security implications.

The study (Qammar et al. 2023) examined chatbot security flaws, attacks, and how malicious actors take advantage of these platforms, with a special emphasis on ChatGPT. Addressing of the growing cybersecurity issues brought about by chatbots' transition to more complex AI models like ChatGPT. Examines past chatbot advancements, cybersecurity risks, and particular ChatGPT vulnerabilities, along with case studies of abusive use. demonstrates how ChatGPT may be abused to send phishing emails, malware, and other online threats. Acknowledges the limitations of the quickly changing nature of AI technologies in terms of anticipating and mitigating any potential security concern.

Shoaib et al. (2023), This study explores the challenges of deepfakes and misinformation enabled by advanced AI, focusing on large models and proposing defense strategies. It highlights societal impacts, the effectiveness of detection algorithms, and the need for user education, ethical oversight, and collaborative approaches. While an integrated framework is proposed, limitations include the rapid evolution of AI, potential algorithmic shortcomings, and global cooperation hurdles. Future research address these gaps through behavioral insights, economic models, and innovative technologies.

In Qammar et al. (2023), the study examined the impact of ChatGPT in cyberspace, focusing on its role in cybersecurity and potential threats. Addressed the dual-use of ChatGPT in cybersecurity, posing both opportunities for defense and risks for exploitation. Analyzed the functionality of ChatGPT, cybersecurity case studies, and explores potential misuse scenarios. Highlighted the versatile use of ChatGPT in creating malware, phishing emails, and its role in cyber defense strategies. The study noted the challenges in fully predicting and mitigating the evolving threats posed by AI technologies in cybersecurity.

In Kaheh et al. (2023), the study investigates the use of Large Language Models (LLMs), like GPT-4, in cybersecurity through a conversational agent called Cyber Sentinel, designed to streamline security tasks. It integrates components like an IoC database, SIEM, and GPT-4, using prompt engineering for task automation. While experiments show LLMs' potential for cybersecurity, limitations include reliance on human judgment for complex threats, challenges in operational effectiveness, and privacy and ethical concerns. The study emphasizes

the need for careful management of these issues to mitigate risks and enhance the applicability of AI in cybersecurity.

In Alawida et al. (2023), the study examines the misuse of ChatGPT in cyberattacks, highlighting its vulnerability to exploitation by malicious actors posing as regular users. Using a survey methodology, it investigates tactics used to manipulate ChatGPT, the associated cybersecurity threats, and user awareness of these risks. Results reveal the potential for ChatGPT to be leveraged in cyberattacks and emphasize the need for greater awareness and mitigation strategies to address these vulnerabilities.

In Botacin (2023), The study investigates the potential misuse of GPT-3 for generating malware, focusing on its ability to create or modify code through completion and edit modes. Experiments reveal that while GPT-3 can generate malware components from descriptions and apply obfuscation techniques, it struggles to create complete malware samples or fully utilize third-party libraries. The research highlights GPT-3's capabilities in supporting common functions but notes limitations in its context comprehension and the practicality of training it for specialized tasks like obfuscation. These findings underscore concerns about GPT-3 lowering barriers for cyberattacks.

Khoury et al. (2023) aimed to evaluate the security of programs generated by ChatGPT. It explored that ChatGPT can be guided to improve the security of its code output through appropriate prompts and discusses the ethical aspects of using AI for code generation. The study focused on the need to understand the safety of programs generated by ChatGPT, as it can translate natural language into code The main issue addressed that the potential insecurity of code produced by ChatGPT. Despite its advanced capabilities in processing human-like text and translating natural language into code, there are concerns about whether the generated code is robust against various types of cyber attacks. The study asked ChatGPT to generate 21 programs in various programming languages, each designed to highlight specific vulnerabilities (e.g., SQL injection, memory corruption). The instructions to ChatGPT did not specifically demand secure code, simulating the behavior of a novice programmer. The aim was to assess the model's ability to create code resistant to common cyber attacks without explicit instructions for security features.

Collaborative defense strategies for robust threat detection and mitigation are shown in Table 2.

5.1 Casestudies: GenAI generated malwares

- BlackMamba is an advanced polymorphic keylogger developed using OpenAI's GPT-3, showcasing the potential of GenAI to automate the creation of sophisticated malware. This keylogger dynamically generates malicious code in realtime, continuously rewriting its structure during execution to evade traditional antivirus detection methods. By altering its code with each iteration, BlackMamba effectively bypasses signature-based detection techniques employed by most cybersecurity tools. This capability not only complicates detection but also highlights the growing challenge of addressing AI-driven threats in cybersecurity. The emergence of BlackMamba underscores the risks posed by GenAI when misused, emphasizing the need for robust defensive measures to counteract such threats (Intelligence 2023).
- The use of AI-driven frameworks for generating polymorphic ransomware by cyber-

Table 2 Collaborative defense strategies for robust threat detection and mitigation

Challenges	Description	Future directions/recommendations
Paradigm Shift (Shen 2023; Sun et al. 2023)	Represents a departure from static signature-based methods, embracing real-time monitoring of software behavior (Shen 2023).	Shifts focus from predetermined patterns to dynamic actions, adapting to the evolving tactics of sophisticated cyber threats (Sun et al. 2023).
Adaptability	Adapts to the dynamic nature of cyber threats, providing a defense strategy capable of detecting novel and evolving threats (Safitra et al. 2023).	Crucial for staying ahead of advanced and polymorphic malware that may bypass traditional signature-based detection systems (Gopinath and Sethuraman 2023).
Real-time Monitoring	Involves continuous tracking of file accesses, system calls, network communications, and other behavioral attributes during runtime execution (Blair et al. 2023).	Enables the identification of anomalies and malicious activities as they occur, facilitating proactive threat detection and response (Pissanidis and Demertzis 2023).
Integration of Technologies	Explores advanced techniques, including machine learning algorithms, to enhance the accuracy and efficiency of behavioral analysis (Peddireddy and Peddireddy 2023).	Incorporates cutting-edge technologies to improve the precision of threat detection and mitigate the challenges posed by evolving cyber threats (Azam et al. 2023).

criminal groups was found in 2022. These frameworks automated the mutation of ransomware payloads, altering encryption methods, communication protocols, and file structures to evade detection. The integration of GenAI in these frameworks allowed attackers to scale their operations, targeting diverse systems with custom-tailored payloads. Such advancements demonstrated the increasing sophistication of ransomware attacks facilitated by GenAI, emphasizing the need for adaptive and AI-powered defensive systems (Lab 2022).

- DeepLocker, developed by IBM researchers in 2018, serves as an early example of AI-powered malware that adapts dynamically to avoid detection. Although not created using modern GenAI tools, DeepLocker used AI to embed malicious payloads within seemingly benign applications, activating only under specific conditions, such as when targeting a specific individual or environment. This proof-of-concept demonstrated how AI could be weaponized to enhance the stealth and precision of malware. DeepLocker's principles have since inspired further developments in the use of GenAI for creating highly targeted and undetectable threats (Research 2018).

6 Critical analysis and discussions

GenAI is revolutionizing the cybersecurity landscape by automating threat detection and response with its advanced processing and predictive capabilities. These tools not only enhance security protocols but also introduce new challenges in safeguarding digital assets against sophisticated cyber threats. This investigation leads to the comparative analysis of various GenAI tools such as GPT3.5, Scribe, AlphaCode, GitHub Copilot, GPT-4, Bard, Cohere Generate, and Dall-E2 in terms of potential exploitations by evildoers (Tabulated in Table 3).

GPT 3.5: While it is useful for threat detection and predictive insights, it also poses risks in automated attacks and obfuscation. This dual nature makes it a potent tool but also a

Table 3 Qualitative analysis of GenAI tools

Tool	Usefulness for cyber experts	Potential for malicious use
GPT 3.5	- Threat detection: Learns from patterns in cyber threats and vulnerabilities. - Predictive insights: Identifies trends and predicts future threats.	- Automated Attacks: generate sophisticated attack vectors. - Obfuscation:create deceptive content to evade detection.
Scribe	- Rapid querying: Learns documentation of security products, aiding analysts in querying security tools.	- Automated exploits: generate malicious code snippets or exploit payloads. - False documentation: create misleading documentation.
AlphaCode	- Quick solutions: Helps programmers find solutions rapidly.	- Malware Generation: create custom malware variants. - Code injection: generate malicious code.
GitHub copilot	- Productivity boost: Reduces boilerplate code and supports multiple languages.	- Automated attacks: assist in creating malicious scripts or backdoors. - Plagiarism: copy existing malicious code.
GPT-4	- Advanced text analysis: Improved accuracy in threat intelligence.	- Phishing: generate convincing phishing emails or messages. - Disinformation: create misleading content.
Bard	- Artistic exploration: inspire novel approaches to threat analysis.	- Social Engineering: generate persuasive narratives for social engineering attacks.
Cohere generate	- Fine-Tuning: Enables tailored solutions for specific tasks.	- Customized attacks: create targeted attack content. - Misinformation: May generate false information.
Dall-E2	- Visual representation: aid in visualizing threat scenarios.	- Image-Based Attacks: generate malicious images or visual content.

potential threat in the wrong hands. To prevent the malicious use of GPT−3.5 in cybersecurity, real-time content filtering, anomaly detection, and the augmentation of the model with natural language understanding (NLU) capabilities to detect fraudulent content. Adversarial testing assesses the model's resilience to manipulation, while behavioral analysis monitors user interactions for suspicious behavior. Enhancing model explainability and implementing zero trust architecture with strict access controls further mitigate risks, ensuring responsible and secure utilization of GPT−3.5 in cybersecurity contexts as mentioned in Table 4.

Scribe: This tool is valuable for rapid querying, but the potential for automated exploits and false documentation is concerning (mentioned in Table 4). It underscores the need for robust security measures when using such tools. To mitigate the potential for malicious use of Scribe, implement code validation mechanisms, integrating with version control systems for tracking changes, employing automated documentation review processes, enforcing user authentication and access controls, and providing comprehensive training and education to

Table 4 Quantitative analysis of GenAI tools

Tool	Usefulness for cyber experts (%)	Potential for malicious use (%)
GPT- 3.5	70	80
Scribe	70	60
AlphaCode	80	70
GitHub Copilot	90	90
GPT-4	80	60
Bard	60	70
Cohere Generate	80	80
Dall-E2	70	60

users. These measures help ensure the accuracy and integrity of code documentation while reducing the risk of unintentional misinformation or obfuscation, thus preserving Scribe's usefulness for cyber experts while maintaining a balanced risk profile.

AlphaCode: While it provides quick solutions for programmers, the risk of malware generation and code injection is high (mentioned in Table 4). This highlights the importance of using such tools responsibly and ethically. Implement a code sanitization process within AlphaCode to sanitize generated code snippets and remove potentially dangerous or insecure elements. Utilize whitelisting techniques to define a set of approved libraries, functions, and code patterns that meet security standards. Only suggest code snippets that adhere to these predefined criteria, reducing the likelihood of introducing vulnerabilities into the application codebase. Additionally, integrate code review checkpoints to ensure that sanitized code snippets undergo manual inspection by cybersecurity experts before being integrated into production code, further enhancing security controls and reducing the risk of exploitation.

GitHub Copilot: While it enhances productivity, the possibility of automated attacks and plagiarism is a concern. This suggests that while AI can boost efficiency, it can also be misused if not properly regulated (mentioned in Table 4).

GPT-4: It offers improved text analysis, but the risk of phishing and disinformation is present. This underscores the need for vigilance and critical thinking when interpreting AI-generated content (mentioned in Table 4).

Bard: While it inspires novel approaches, social engineering attacks may be generated. This highlights the potential misuse of AI in manipulating human behavior (mentioned in Table 4).

Cohere generate: While tailored solutions are beneficial, the risk of customized attacks and misinformation exists. This suggests the need for stringent security measures and fact-checking.

Dall-E2: It is useful for visual representation, but image-based attacks are a concern. This highlights the need for robust image analysis tools to detect and mitigate such threats (mentioned in Table 4).

To safeguard against malicious use of GitHub Copilot, GPT-4, Bard, Cohere Generate, and Dall-E2, implement integrating static analysis tools, establishing code review workflows, enhancing contextual understanding, enforcing user authentication, and setting API rate limits. Additionally, implementing anomaly detection, input validation, content scanning, and continuous monitoring further fortifies security measures. These measures col-

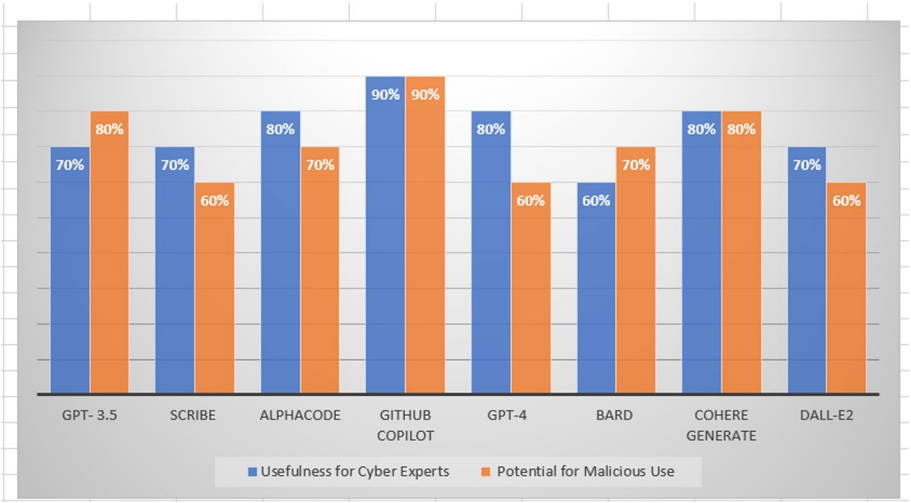

Fig. 4 GenAI tools comparative analysis

lectively mitigate the risk of introducing vulnerabilities or facilitating automated attacks while optimizing the productivity benefits of these GenAI tools for software development.

These AI tools offer significant benefits, cybersecurity professionals need to be cautious of their potential misuse. The dual nature of these tools underscores the importance of ethical AI practices and robust security measures. It is crucial to balance the benefits of AI with the potential risks it poses in the realm of cybersecurity.

Table 4 depicts that the average usefulness of these tools for cyber experts is approximately 74.44%, while the average potential for malicious use is around 72.22%. This comparative analysis indicates that while these AI tools have significant usefulness for cyber experts, they also have a substantial potential for malicious use. The tools with the highest usefulness for cyber experts are GitHub Copilot and Cohere Generate, both scoring 90% and 80% respectively. However, these tools also have a high potential for malicious use, with GitHub Copilot scoring the highest at 90%.

Figure 4 shows the graph of comparison between offensive and defensive side of GenAI. It concludes that GenAI exhibits a moderate level of usefulness for cyber experts, with a 70% rating. This indicates that it offers substantial benefits, particularly in threat detection and predictive insights. However, its potential for malicious use is notably high, rated at 80%. This suggests that while GenAI can enhance cybersecurity efforts, there are significant risks associated with its capacity to autonomously generate sophisticated malware and deceptive content.

While our study provides a snapshot of the current state, ongoing developments in cyber threats may alter the usefulness and risks associated with GenAI tools. Future research endeavors should aim to track GenAI's adaptability to emerging threats to ensure the continued relevance of our findings. Context dependency is another crucial consideration in our analysis. The usefulness and risk profiles of GenAI tools are heavily influenced by contextual factors such as the specific task at hand and the expertise of the user. While our study offers a high-level overview, it may not fully account for the complex variations in different scenarios, highlighting the need for a more granular examination in future research.

6.1 Recommendations

Based on the findings of this study, several recommendations can be made to address the challenges posed by GenAI in cybersecurity:

Policies must differentiate between high-risk and low-risk use cases of GenAI, especially when it comes to code generation. For instance, the generation of encryption and decryption code using GenAI models can pose significant risks. When modified, such code can be adapted to create ransomware or other malicious software, leading to severe cybersecurity threats. To mitigate these risks, it is recommended that a mandatory licensing process be implemented for all GenAI tools used for code generation. This process would ensure that AI systems used in sensitive areas, like cryptographic code development, undergo strict vetting and monitoring.

Furthermore, high-risk use cases, such as those involving the generation of potentially malicious code (e.g., malware or ransomware), should be subject to enhanced regulatory scrutiny. These cases should be continuously monitored, and access to such functionalities should be restricted to users with verified and legitimate purposes. By refining the process of code generation and applying licensing and monitoring, it becomes possible to prevent the misuse of GenAI tools in creating harmful software. This regulatory approach would ensure that such technologies are used responsibly, reducing the likelihood of malicious exploitation while enabling beneficial use cases.

Hence, GenAI represents a paradigm shift with vast potential, its dual-use nature necessitates cautious and constructive approaches. By combining technological safeguards with forward-thinking regulatory policies, we can harness the benefits of GenAI while minimizing its risks. The complexities of navigating this rapidly evolving technology require both vigilance and a clear vision for the future.

7 Conclusion

The investigation into the cyber threats posed by GenAI in malware generation reveals a dual-edged reality. While GenAI demonstrates significant utility in cybersecurity, particularly in threat detection and predictive analysis, it also presents considerable risks, particularly in the autonomous creation of sophisticated malware. Its usefulness rate is about 70%, while the likelihood of malicious exploitation is estimated at 80%, highlighting the need for careful regulation.

The research stresses the importance of developing policies and frameworks to address these risks while allowing GenAI to be beneficial in cybersecurity. A particular concern is the generation of sensitive code, such as encryption and decryption algorithms, which could be repurposed into malicious software like ransomware. To mitigate these threats, the study suggests mandatory licensing and regulatory oversight for AI systems involved in high-risk applications, such as code development. By implementing thorough evaluation processes and continuous monitoring, the potential for misuse can be reduced while still fostering innovation.

Additionally, high-risk functionalities, such as malware generation, should be limited to verified users with legitimate purposes, subject to enhanced regulatory oversight. This would prevent malicious exploitation while ensuring the responsible use of GenAI. A bal-

anced approach, distinguishing between high-risk and low-risk applications, would allow policymakers to ensure responsible deployment of GenAI technologies, preserving their transformative potential for cybersecurity without compromising security.

In conclusion, the study emphasizes the need for a proactive, refined strategy to navigate the complex interplay between the benefits and risks of GenAI. This approach can unlock its potential to enhance cybersecurity resilience while mitigating its vulnerability to malicious exploitation, ultimately achieving a balance between innovation and security.

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Code availability Not applicable.

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